
FreshBasket AWS Cloud Migration

Cloud Architecture & Deployment Report

Subject 42904 — Cloud Computing and Software as a Service
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PART 1 — AWS SYSTEM ARCHITECTURE (Deliverable 1)

1.1 Architecture Overview

The proposed architecture transitions FreshBasket from a single desktop PC, a critical single point of failure, to a robust, scalable cloud environment on Amazon Web Services. The design is multi-tier and highly available, centred on AWS Elastic Beanstalk for application orchestration and Amazon RDS for managed data persistence.

Incoming HTTP traffic enters through an Application Load Balancer (ALB), which distributes requests across a fleet of EC2 instances managed by an Auto Scaling Group (ASG). Instances are spread across two distinct Availability Zones within a custom Virtual Private Cloud (VPC), guaranteeing fault tolerance at the compute tier. The application communicates with an Amazon RDS MySQL database in Multi-AZ mode, providing synchronous replication and automatic failover. Amazon SNS delivers email alerts for environment health changes and scaling events.

1.2 Architecture Diagram

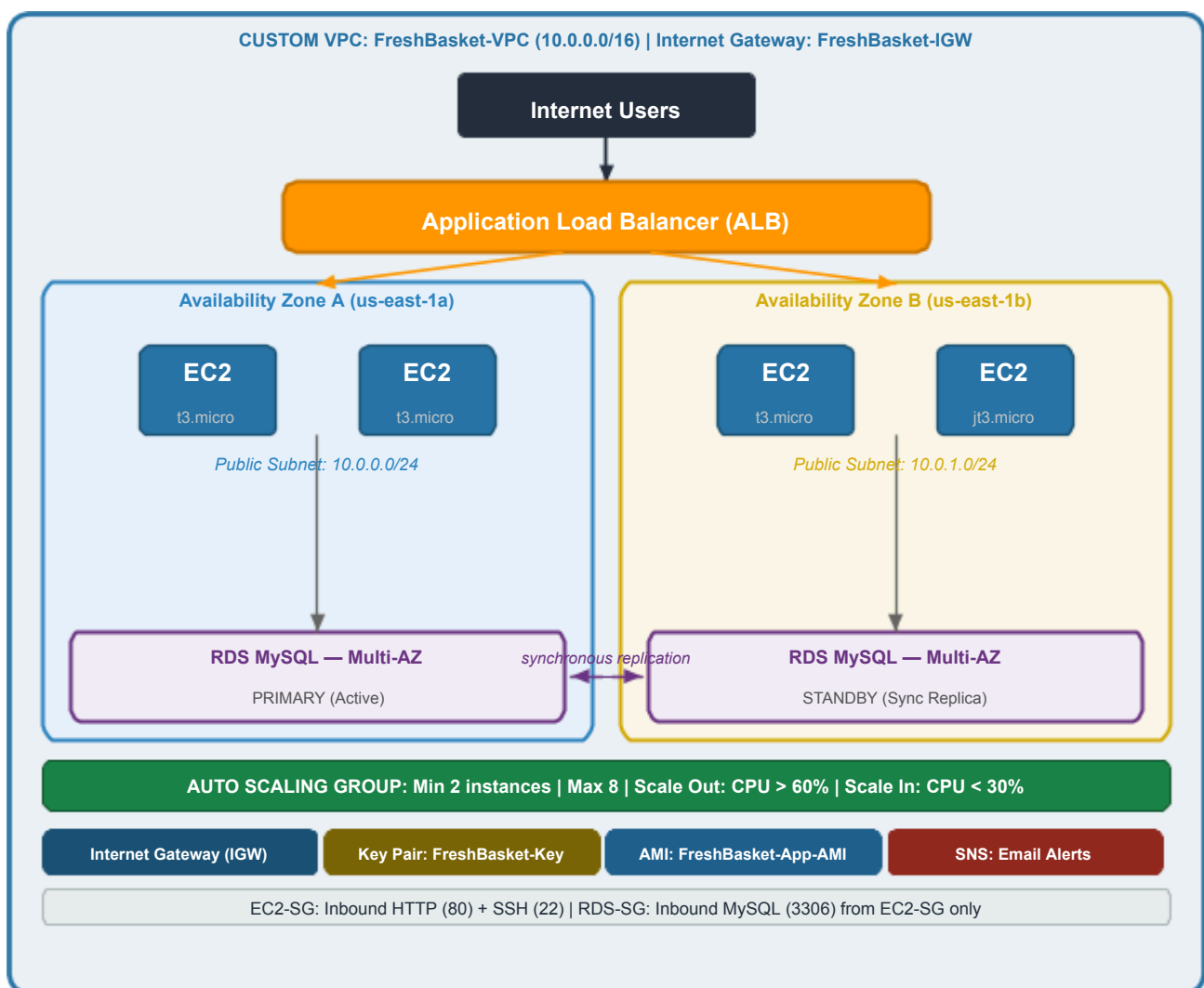


Figure 0: FreshBasket AWS Architecture — Internet Users → ALB → EC2 Auto Scaling Group (min 2, max 8 across 2 AZs) → RDS MySQL Multi-AZ (synchronous replication, automatic failover).

1.3 Justification of AWS Components

Component	Scalability Justification	Disaster Recovery Justification
Custom VPC (10.0.0.0/16)	/16 CIDR provides 65,536 IPs — ample room for subnet growth as the business expands.	Multi-AZ subnets ensure network-level redundancy; an AZ outage does not affect the other subnet.
Application Load Balancer	Auto-scales its own capacity; distributes requests evenly across all healthy EC2 instances.	Continuous health checks on EC2 targets; routes traffic only to healthy instances across both AZs.
Auto Scaling Group (2–8)	Adds instances when CPU > 60%, removes when < 30% — directly addresses the TV-surge risk without over-provisioning.	Maintains minimum 2 instances across 2 AZs at all times; auto-replaces any failed instance.
Custom AMI (FreshBask et-App-AMI)	Pre-installed dependencies reduce instance boot time during scale-out, enabling faster response to traffic spikes.	Every ASG-launched instance is identically configured — eliminates drift and ensures consistent behaviour.
Amazon RDS MySQL (Multi-AZ)	Managed service handles storage auto-scaling and IOPS; read replicas can be added for future read-heavy loads.	Synchronous standby replica in a separate AZ; automatic failover in 60–120 s with zero data loss.
EC2 Security Group	Stateful firewall scales automatically with the ASG — no manual reconfiguration as instances are added.	Restricts inbound to ports 80 (HTTP) and 22 (SSH) only, minimising the attack surface.
RDS Security Group	Scales with the database tier without manual intervention.	MySQL (3306) access restricted exclusively to the EC2 SG — database never reachable from the internet.
Amazon SNS (Notifications)	Alerts operators to scaling events, enabling proactive capacity planning.	Immediate health-change notifications allow rapid human response to issues automated recovery cannot resolve.

1.4 Statement of Assumptions

Category	Details
AWS Region	All resources are deployed in ap-southeast-2 (Sydney) to minimise latency for FreshBasket's Sydney-based user base.
Instance Type	EC2 instances use t3.micro — CPU-burstable, suitable for variable web traffic.
Database Instance	RDS uses db.t3.small with 20 GB allocated storage and MySQL 8.0.
Application Architecture	The web application is stateless. Session data is stored client-side (cookies) or in RDS, so any EC2 instance can serve any request.
Security Strategy	The database requires no public internet access. All database interactions occur through the EC2 application tier. SSH (port 22) is open from any IP for development.
IAM Constraints	The AWS Academy Learner Lab provides two pre-built identities: LabRole (Beanstalk service role) and LabInstanceProfile (EC2 instance profile). No custom IAM roles can be created.
Key Pair	A single RSA key pair (FreshBasket-Key) is shared across all EC2 instances for SSH access.
DB Credentials	RDS connection details are injected via Elastic Beanstalk environment properties: RDS_HOSTNAME, RDS_PORT, RDS_DB_NAME, RDS_USERNAME, RDS_PASSWORD.

1.5 Complete List of AWS Services Used

#	AWS Service	Purpose in Architecture
1	Amazon VPC	Custom isolated network (10.0.0.0/16) with two public subnets
2	Internet Gateway	Provides internet connectivity to VPC resources
3	Route Table	Routes 0.0.0.0/0 traffic to the Internet Gateway for both public subnets
4	AWS Elastic Beanstalk	PaaS orchestration of the full application environment (EC2, ALB, ASG)
5	Amazon EC2	Compute instances running the PHP web application
6	Custom AMI	Pre-configured image (FreshBasket-App-AMI) for consistent, fast instance launches
7	EC2 Key Pair	RSA key pair (FreshBasket-Key) for SSH authentication to all instances
8	Application Load Balancer	HTTP traffic distribution across EC2 instances in both Availability Zones
9	Auto Scaling Group	Dynamic capacity management — 2 to 8 instances, CPU-based scaling triggers
10	Amazon RDS (MySQL)	Managed relational database with Multi-AZ deployment for high availability
11	Security Groups	Stateful firewalls for EC2 (HTTP/SSH) and RDS (MySQL from EC2-SG only)
12	Amazon SNS	Email notifications for environment health changes and scaling events

PART 2 — AWS IMPLEMENTATION EVIDENCE (Deliverable 2)

Figure 1. Custom VPC

VPC overview showing VPC name FreshBasket-VPC, CIDR 10.0.0.0/16, two subnets in different AZs, and route table with 0.0.0.0/0 routed to the Internet Gateway.

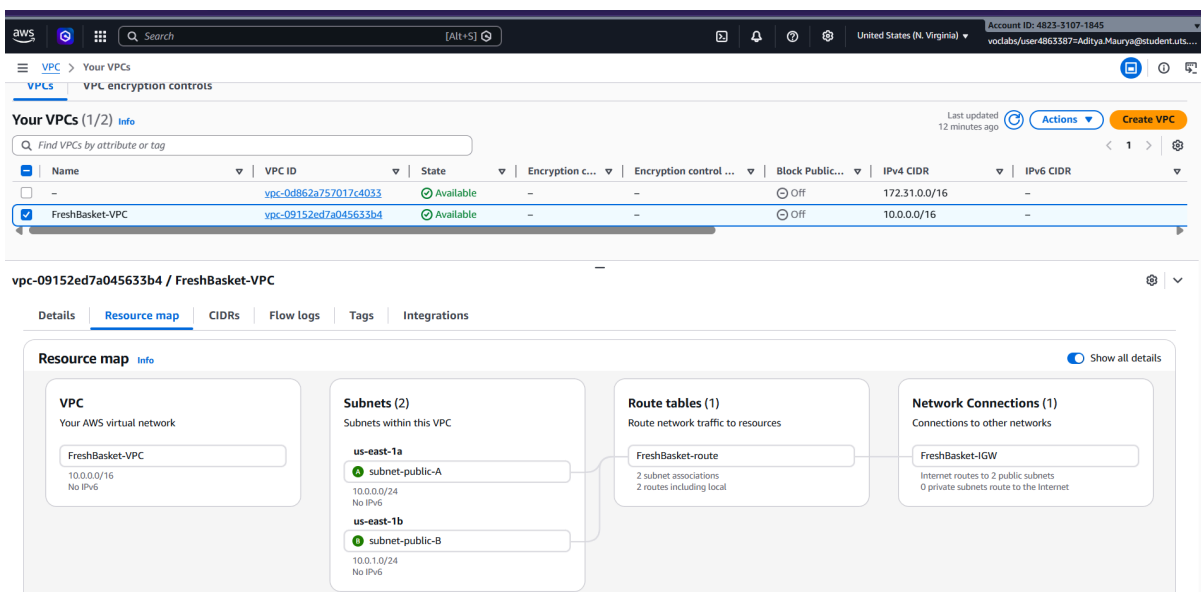


Figure 2. Route Table

Route table view confirming that both subnets are public (with a default route to an Internet Gateway)

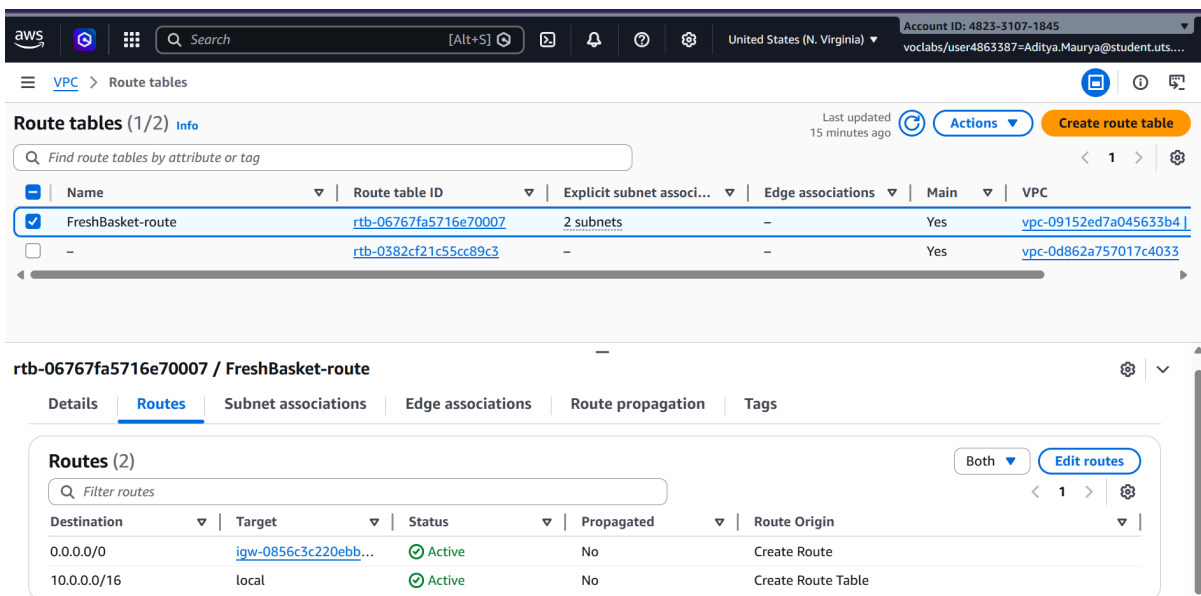


Figure 3. EC2 Security Group

Inbound rules of the FreshBasket-SG showing port 22 (SSH) and port 80 (HTTP) open from 0.0.0.0/0.

The screenshot displays the AWS Management Console interface for the 'Security Groups' section. A green notification banner at the top states: 'Inbound security group rules successfully modified on security group (sg-00b583ab287e27717 | FreshBasket-SG)'. Below this, the 'Security Groups (1/3)' list shows three groups: 'FreshBasket-SG' (sg-00b583ab287e27717), 'default' (sg-00543e9a8d898eb77), and another 'default' (sg-00826313b64a64b4d). The 'FreshBasket-SG' group is selected, and its 'Inbound rules' tab is active. The 'Inbound rules (2)' table lists two rules:

Name	Security group rule ID	IP version	Type	Protocol	Port range	Source	Description
-	sg-0f5b1897d8add6b5c	IPv4	HTTP	TCP	80	0.0.0.0/0	allow web traffic
-	sg-044405baf3b233675	IPv4	SSH	TCP	22	0.0.0.0/0	allow SSH access

Figure 4. RDS Group

Inbound rules of FreshBasket-DB-SG showing port 3306 (MySQL) with source set to the EC2 security group ID — not a CIDR range.

The screenshot displays the AWS Management Console interface for the 'Security Groups' section. The 'Security Groups (1/4)' list shows four groups: 'FreshBasket-DB-SG' (sg-06151039a90845a8c), 'default' (sg-00543e9a8d898eb77), 'default' (sg-00826313b64a64b4d), and 'FreshBasket-SG' (sg-00b583ab287e27717). The 'FreshBasket-DB-SG' group is selected, and its 'Inbound rules' tab is active. The 'Inbound rules (1)' table lists one rule:

Name	Security group rule ID	IP version	Type	Protocol	Port range	Source	Description
-	sg-073cd409be52f6032	-	MySQL/Aurora	TCP	3306	sg-00b583ab287e2771...	allow MySQL from app ...

Figure 5. Custom EC2 Key Pair

EC2 → Key Pairs page showing the custom key pair FreshBasket-Key created for this deployment.

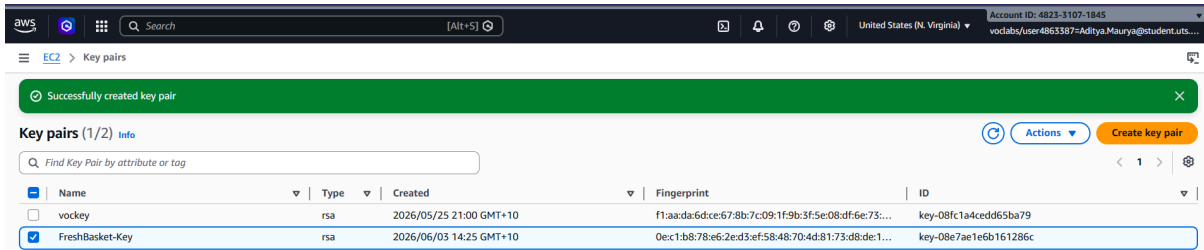


Figure 6. Custom AMI

EC2 → AMIs page (filter: Owned by me) showing FreshBasket-App-AMI with status Available.

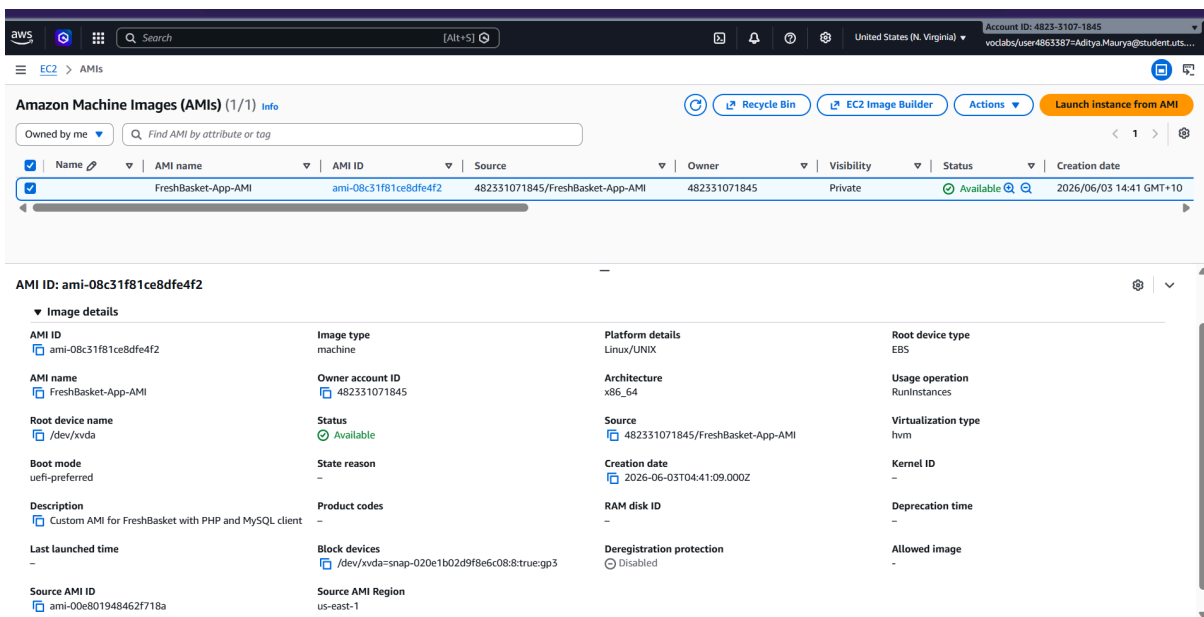


Figure 7. RDS Multi AZ Configuration

RDS instance summary page with Multi-AZ field clearly showing Yes and the two Availability Zones in use.

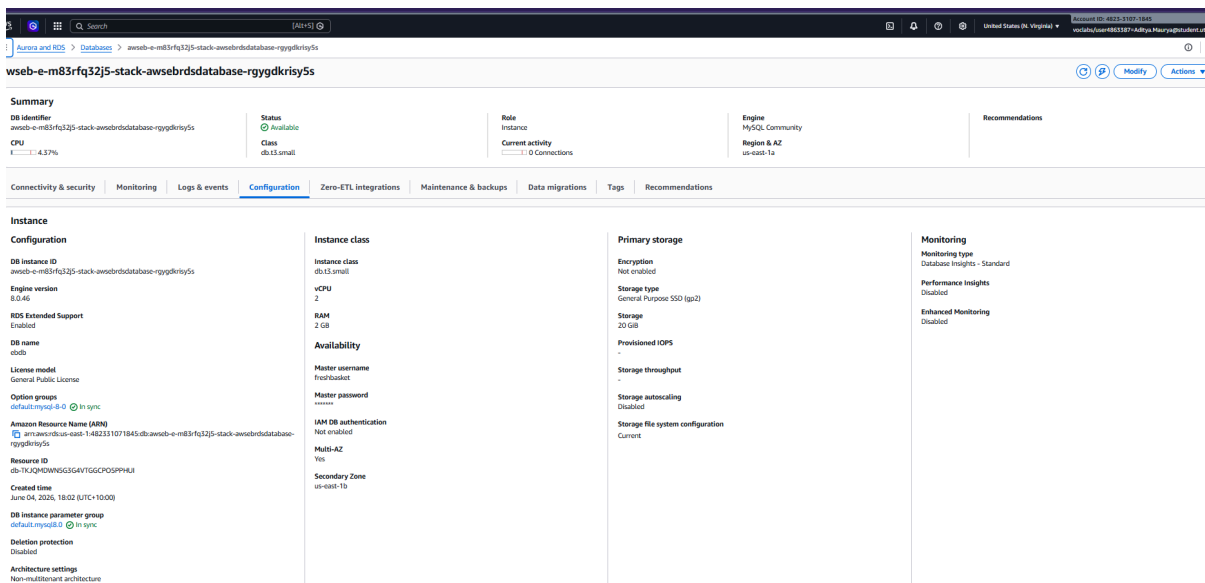


Figure 8. Elastic Beanstalk Dashboard

Beanstalk environment overview showing health status Ok and the environment URL.

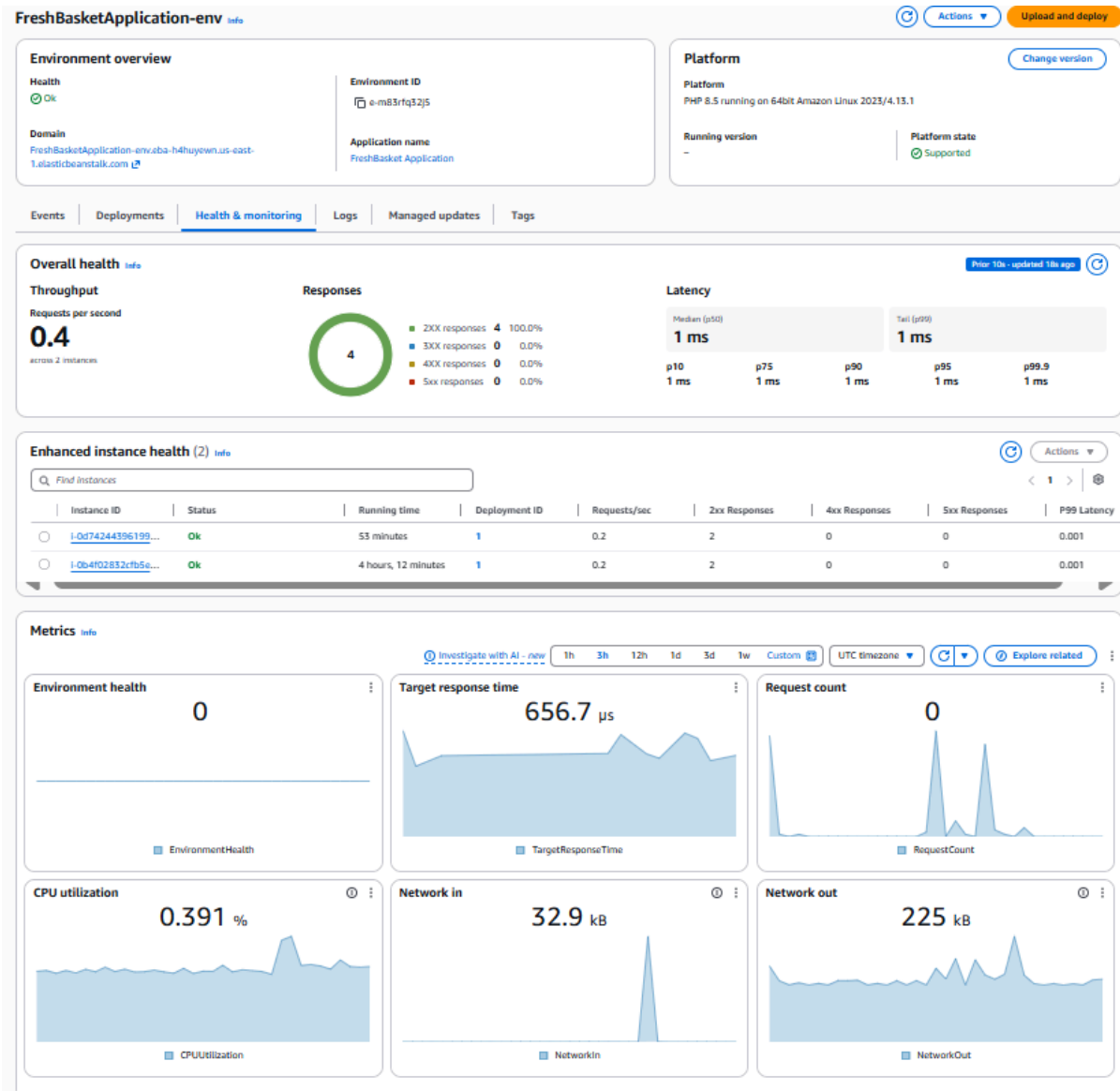


Figure 9. Configuration Panel

Configuration panel with LabRole (Service role) and LabInstanceProfile (EC2 instance profile) visible

The screenshot shows the AWS Elastic Beanstalk Configuration Panel for the 'FreshBasketApplication-env' environment. The 'Service access' section is expanded, showing the 'Service role' as 'arn:aws:iam::482331071845:role/LabRole', the 'EC2 key pair' as 'FreshBasket-Key', and the 'EC2 instance profile' as 'LabInstanceProfile'. The 'Networking and database' section is also expanded, showing VPC settings, instance subnets, and database configuration. The 'Database' section includes fields for 'Database availability', 'Database engine', 'Database password', 'Database username', 'Has coupled database', 'Database engine version', 'Database deletion policy', 'Database instance class', and 'Database subnets'.

Service access		
Service role	EC2 key pair	EC2 instance profile
arn:aws:iam::482331071845:role/LabRole	FreshBasket-Key	LabInstanceProfile

Networking and database		
VPC	Instance subnets	
vpc-09152ed7a045633b4	subnet-092bbe410a12abd1a,subnet-062d30e99e3253225	
Database		
Database availability	Has coupled database	Database deletion policy
true	true	Delete
Database engine	Database engine version	Database instance class
mysql	8.0.46	db.t3.small
Database password	Database storage	Database subnets
*****	20	subnet-092bbe410a12abd1a,subnet-062d30e99e3253225
Database username		
freshbasket		

Figure 10. Auto Scaling Configuration

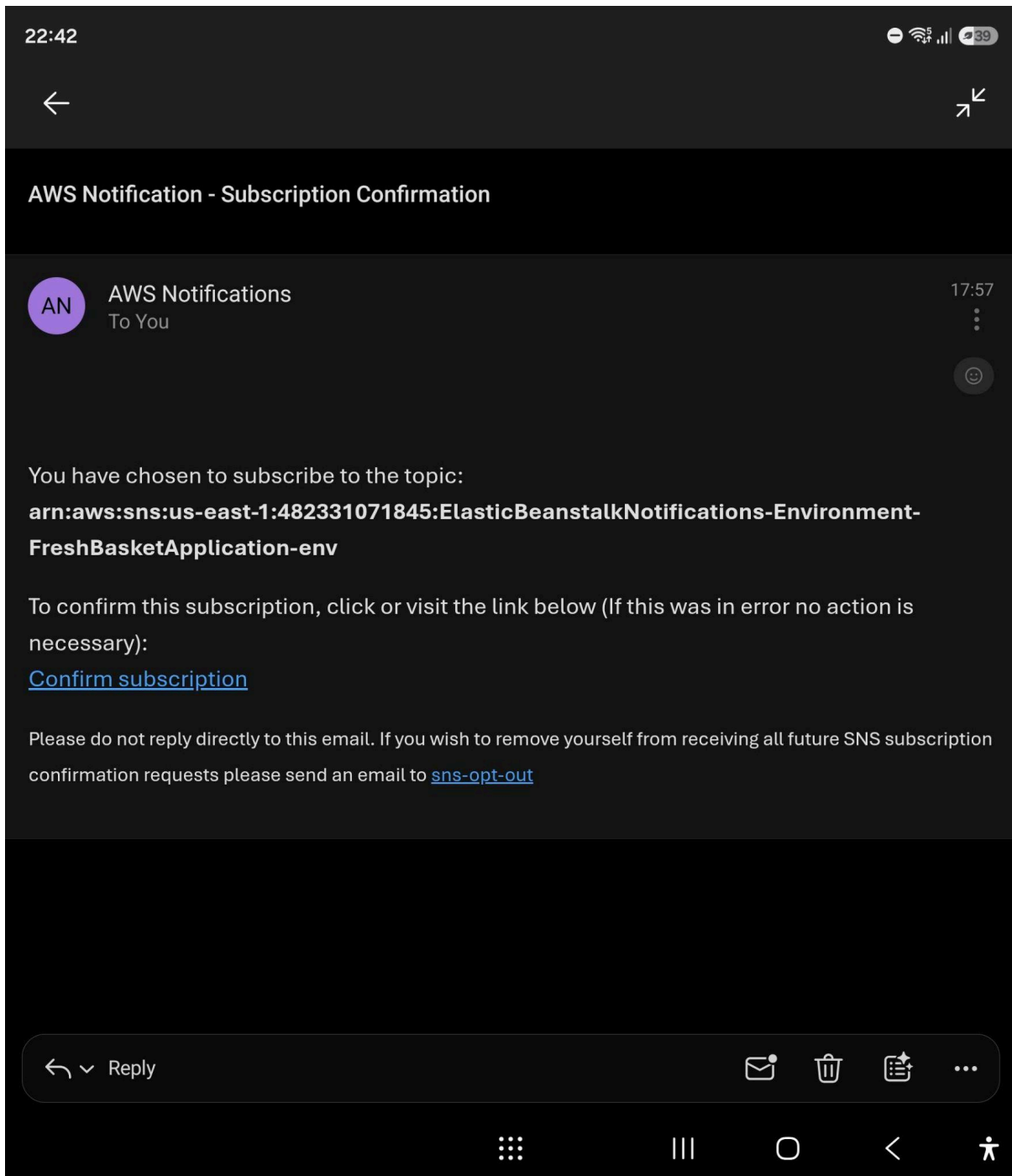
Beanstalk Configuration → Capacity showing min 2, max 8, CPU scale-out at 60% and scale-in at 30%.

The screenshot shows the AWS Elastic Beanstalk Configuration Panel for the 'FreshBasketApplication-env' environment, specifically the 'Instance traffic and scaling' section. This section is used to customize the capacity and scaling for the environment's instances. It includes settings for 'Instances' (Root volume type, Instance size, IMDSv1), 'EC2 Security Groups', 'Capacity' (Environment type, Min instances, Max instances, Fleet composition, On-demand base, Capacity rebalancing, Scaling cooldown, Processor type, AMI ID, Availability Zones, Metric, Unit, Period, Upper threshold, Scale up increment, Scale down increment), 'Load balancer' (Load balancer visibility, Load balancer is shared, Load balancer subnets, Store logs, Load balancer type, IP address type).

Instance traffic and scaling		
Instances		
Root volume type	Instance size	IMDSv1
gp2	20	Disabled
EC2 Security Groups		
sg-00b583ab287e27717,sg-069798e3ff7d1d3a0		
Capacity		
Environment type	Min instances	Max instances
Load balanced	2	8
Fleet composition	On-demand base	On-demand above base
On-Demand instances	0	0
Capacity rebalancing	Scaling cooldown	Processor type
Disabled	360	x86_64
Instance types	AMI ID	Availability Zones
t3.micro, t3.small	ami-0a405e151a9cc0b4a	Any
Metric	Statistic	Unit
CPUUtilization	Average	Percent
Period	Breach duration	Upper threshold
5	5	60
Scale up increment	Lower threshold	Scale down increment
1	30	-1
Load balancer		
Load balancer visibility	Load balancer subnets	Load balancer type
public	subnet-092bbe410a12abd1a,subnet-062d30e99e3253225	application
Load balancer is shared	Store logs	IP address type
false	Disabled	ipv4

Figure 11. Email Notifications

SNS subscription confirmation email AND at least one Elastic Beanstalk event notification email, both with subject and timestamp visible.



22:41

39



AWS Elastic Beanstalk Notification - Environment health has transitioned from Ok to Info.
Applica...



AWS Notifications
To You

22:37



Timestamp: Thu Jun 04 12:36:51 UTC 2026

Message: Environment health has transitioned from Ok to Info. Application restart in progress (running for 3 seconds).

Environment: FreshBasketApplication-env

Application: FreshBasket Application

Environment URL: <http://FreshBasketApplication-env.eba-h4huyewn.us-east-1.elasticbeanstalk.com>

NotificationProcessId: b22d6238-42db-4199-844d-f797902d40bf

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<https://sns.us-east-1.amazonaws.com/unsubscribe.html?SubscriptionArn=arn:aws:sns:us-east-1:482331071845:ElasticBeanstalkNotifications-Environment-FreshBasketApplication-env:2d4eb886-fd93-478d-a346-d784a90b55dc&Endpoint=aditya.maurya@student.uts.edu.au>

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